LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the official syllabus sentence (quadratics with real coefficients), power-sum recurrences and closed forms for $\alpha^n + \beta^n$ are extension theory. Nothing here is needed for Part I; the habit — never solve for roots when a symmetric combination is asked — is the transferable skill.

Power sums from coefficients alone

SHORT DERIVATION

Newton's recurrence for two roots

α, β satisfy $ax^2 + bx + c = 0$. Multiply the equation for α by α^{n-2}, the one for β by β^{n-2}, and add. With $S_n = \alpha^n + \beta^n$:

$$aS_n + bS_{n-1} + cS_{n-2} = 0, \qquad S_0 = 2, \quad S_1 = -\tfrac{b}{a}$$

So $S_n = \frac{-bS_{n-1} - cS_{n-2}}{a}$ generates every power sum from the coefficients — no radicals, ever. Start at $S_0 = 2$, not S_1 (checklist item).

WORKED EXAMPLE

Worked — Newton's sums

$x^2 - 5x + 3 = 0$: $S_0 = 2$, $S_1 = 5$;
$S_2 = 5 \cdot 5 - 3 \cdot 2 = 19$;
$S_3 = 5 \cdot 19 - 3 \cdot 5 = 80$. Matches $S^2 - 2P = 19$ and $S^3 - 3PS = 125 - 45 = 80$ ✓ (and numerically $\alpha \approx 4.3028, \beta \approx 0.6972$ give $\alpha^3 + \beta^3 \approx 80.00$ ✓).

KEY RESULT

The golden quadratic generates Fibonacci and Lucas

For $x^2 - x - 1 = 0$ with roots $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$; induction on $\alpha^2 = \alpha + 1$ gives

$$\alpha^n = F_n\alpha + F_{n-1} \qquad (F_0 = 0, F_1 = 1)$$

for both roots. Adding the two identities: $S_n = \varphi^n + \psi^n = F_n + 2F_{n-1} = F_{n+1} + F_{n-1} =$, the **Lucas numbers** $2, 1, 3, 4, 7, 11, 18, 29, 47, \ldots$ Subtracting and dividing by $\sqrt{5} = \varphi - \psi$ gives **Binet**: $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$. Worked check: $F_{10} = 55$, and $55\varphi + 34 \approx 122.9918 \approx \varphi^{10}$ ✓.

STANDARD APPLICATION

Standard applications

Integer-valuedness: if S, P are integers, every S_n is an integer (recurrence), so $\varphi^n + \psi^n \in \mathbb{Z}$ despite the radicals — the standard trick for proving $\lfloor \varphi^{2n} \rfloor$-type results, using $|\psi| < 1$ so $\psi^n \to 0$ and the integer L_n sits within 1 of φ^n.

COMMON TRAP

Common traps — power sums

- Recurrence sign: $aS_n + bS_{n-1} + cS_{n-2} = 0$ — move terms carefully.
- $S_0 = 2$ (two roots), including when a root is repeated.
- Binet needs $\varphi - \psi = \sqrt{5}$ in the denominator, not $2\sqrt{5}$.

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

The discriminant as $a^2(\alpha - \beta)^2$, rational-root criteria, and the resolvent solution of the cubic are classical extension material beyond the official quadratic-equations syllabus line.

KEY RESULT

What the discriminant actually measures

$$\Delta = b^2 - 4ac = a^2(\alpha - \beta)^2$$

The factor $a^2 > 0$ is pure scale: the **sign** of Δ is the sign of $(\alpha - \beta)^2$ — positive (distinct real), zero (coincident), negative (non-real conjugates, where $\alpha - \beta$ is purely imaginary). The **value** measures root separation. One number, complete information.

KEY RESULT

Rational roots are a perfect-square question

For rational a, b, c: roots rational $\Leftrightarrow$ Δ is the square of a rational. Worked: $x^2 - kx + (k-1) = 0$ has $\Delta = k^2 - 4k + 4 = (k-2)^2$, a perfect square for **every** integer k; roots $x = \frac{k \pm |k-2|}{2} = 1, k-1$ (indeed $x^2 - kx + k - 1 = (x-1)(x-k+1)$ ✓).

SHORT DERIVATION

Lagrange's resolvent — a cubic solved by one quadratic

For $x^3 + px^2 + qx + r = 0$, the shift $x = y - \frac{p}{3}$ kills the square term: depressed cubic $y^3 + Py + Q = 0$. Set $y = u + v$; then $y^3 = u^3 + v^3 + 3uvy$, so matching needs $3uv = -P$ and $u^3 + v^3 = -Q$. Hence $u^3 v^3 = -\frac{P^3}{27}$, and u^3, v^3 are the roots of the **resolvent quadratic**

$$t^2 + Qt - \frac{P^3}{27} = 0$$

Solve it, take cube roots chosen with $uv = -\frac{P}{3}$ (the pairing constraint is essential — arbitrary cube roots give wrong y), and $y = u + v$. Worked: $x^3 - 6x^2 + 11x - 6 = 0$: $x = y + 2$ gives $y^3 - y = 0$; resolvent $t^2 + \frac{1}{27} = 0$; the three admissible (u, v) pairings give $y = 1, 0, -1$, so $x = 1, 2, 3$ ✓ (sum $6 = -p$, product $6 = -r$ ✓).

INTUITION

Why the ladder stops at degree 4

Quadratic → cubic (resolvent quadratic) → quartic (resolvent cubic): each degree is solved by the previous. At degree 5 the ladder provably ends — the general quintic is not solvable by radicals (Abel–Ruffini), and the explanation of *why* is Galois theory. The resolvent is the last rung reachable by pure algebra.

COMMON TRAP

Common traps — resolvents and discriminants

- Taking cube roots of u^3, v^3 independently: the three choices must satisfy $uv = -\frac{P}{3}$.
- "Rational coefficients" hypothesis dropped: Δ a perfect square decides rationality only over $\mathbb{Q}$.
- $\Delta = a^2(\alpha - \beta)^2$, not $(\alpha - \beta)^2$ — the a^2 matters for exact values.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IMO-level)

Vieta jumping is a number-theoretic descent technique, far beyond the JEE syllabus. It is included because its engine is exactly Part I material: Vieta's relations plus root-flipping.

KEY RESULT

The move, isolated

Given a symmetric condition on positive integers (a, b) with a fixed integer quotient k: read the condition as a **quadratic in one variable**. One root is a; Vieta hands you the other root $a' = (\text{sum of roots}) - a = \frac{\text{product}}{a}$, automatically an integer with the same k. If $0 \le a' < b \le a$, the pair (b, a') is a smaller solution with the same k — descend until forced to stop; the terminal configuration reveals what k must be.

SHORT DERIVATION

IMO 1988 Problem 6 — the famous application

Claim: if a, b are positive integers with $ab+1 \mid a^2+b^2$, then $k = \frac{a^2+b^2}{ab+1}$ is a perfect square. Fix such a k and view $a^2 - kba + (b^2 - k) = 0$ as a quadratic in a. The other root is $a' = kb - a = \frac{b^2-k}{a}$; an integer since $a \mid b^2 - k$ (rearrange the equation: $b^2 - k = a(kb - a)$). Ordering $A \geq B$, one checks $0 \leq a' < B$, so descent strictly shrinks the pair while k never changes. It can only halt at $a' = 0$, where the equation gives $k = B^2$ — a perfect square, and the same k as the original pair. ∎

WORKED EXAMPLE

Worked — tracing the descent

$(a, b) = (8, 30)$: $k = \frac{64+900}{241} = 4$. Order $A = 30 \geq B = 8$; other root $a' = 4 \cdot 8 - 30 = 2$, pair $(8, 2)$ (check $\frac{64+4}{17} = 4$ ✓). Reorder $A = 8, B = 2$: $a' = 4 \cdot 2 - 8 = 0$; halt with $k = 2^2 = 4$ ✓. Brute force over $1 \leq a, b < 300$: every admissible quotient is a perfect square ($1, 4, 9, 16, 25, 36, \ldots$) — no exceptions.

STANDARD APPLICATION

The $k = 9$ chain (building solutions up)

Descent runs from any solution down to $(0, 3)$ for $k = 9$; reversing $(a, b) \mapsto (b, kb - a)$ builds all solutions upward: $(0, 3) \to (3, 27) \to (27, 240) \to (240, 2133) \to$ Checks: $\frac{9+729}{82} = 9$ ✓; $\frac{729+57600}{6481} = 9$ ✓. So with $a, b \leq 500$: $(3, 27), (27, 3), (27, 240), (240, 27)$.

COMMON TRAP

Common traps — Vieta jumping

- Forgetting k is *invariant* under the flip — that invariance is the entire proof.
- Assuming a' is a non-negative integer without proof: integrality comes from Vieta (product/sum of roots), non-negativity from $a' = \frac{B^2-k}{A}$ plus the ordering argument; both need care in a written solution.
- Descending without a well-ordering: the descent terminates because positive integers cannot decrease forever — state it.